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3. Drones such as the one shown in the picture have an increasing number of uses in society. This question considers some of the Physics associated with drones.

(a) (i) State Newton's third law of motion. [1]

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(ii) The drone in the picture has four rotors and is shown hovering stationary in the air. The table describes two vertical forces acting on the drone. According to Newton's third law each of these forces is part of a pair of forces. Identify the other force in each case. [2]

Forces acting vertically downwards	Forces acting vertically upwards
Gravitational force of Earth on drone	....
....	Force of air on rotors

(b) The drone is able to hover when each rotor produces a downward flow of air of speed $22\,\text{m}\,\text{s}^{-1}$. The momentum given to the air by the 4 rotors produces an upward force which keeps the drone stationary in the air.

(i) State the relationship between force and momentum. [1]

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(ii) The rate of change of momentum of the air **from the 4 rotors** can be expressed as:

$$4\pi r^2 \rho v^2$$

where r is the radius of each of the rotor blades, v is the speed of the air which is set in motion and ρ is the density of the air. Use this expression to show that the weight of the drone is approximately 20 N. [ρ = 1.3 kg m^{-3}, r = 5.0 × 10^{-2} m] [2]

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(iii) Calculate the vertical acceleration of the drone when v is increased to 24 m s^{-1}. [4]

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(c) (i) Define *velocity*. [1]

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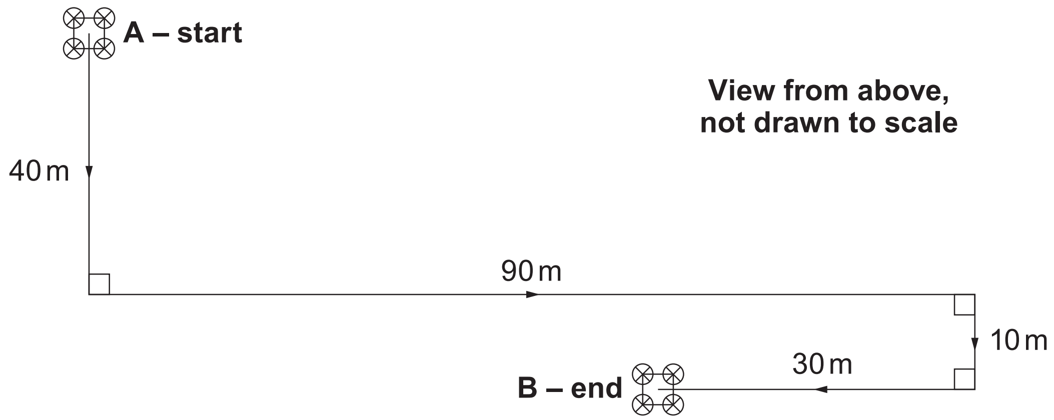
(ii) Bill flies the drone from **A** to **B** in a time of 32 s along the path shown below. The drone is kept at the same height above the ground for the entire flight.



Bill's friend Ted believes that, in flying this path (from **A** to **B**), the mean speed of the drone is more than twice the magnitude of its mean velocity. Bill disagrees. Use the information given to determine who is correct. [4]

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(d) Drones have many applications in today's society. Give **one** benefit and **one** risk of drone technology. [2]

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